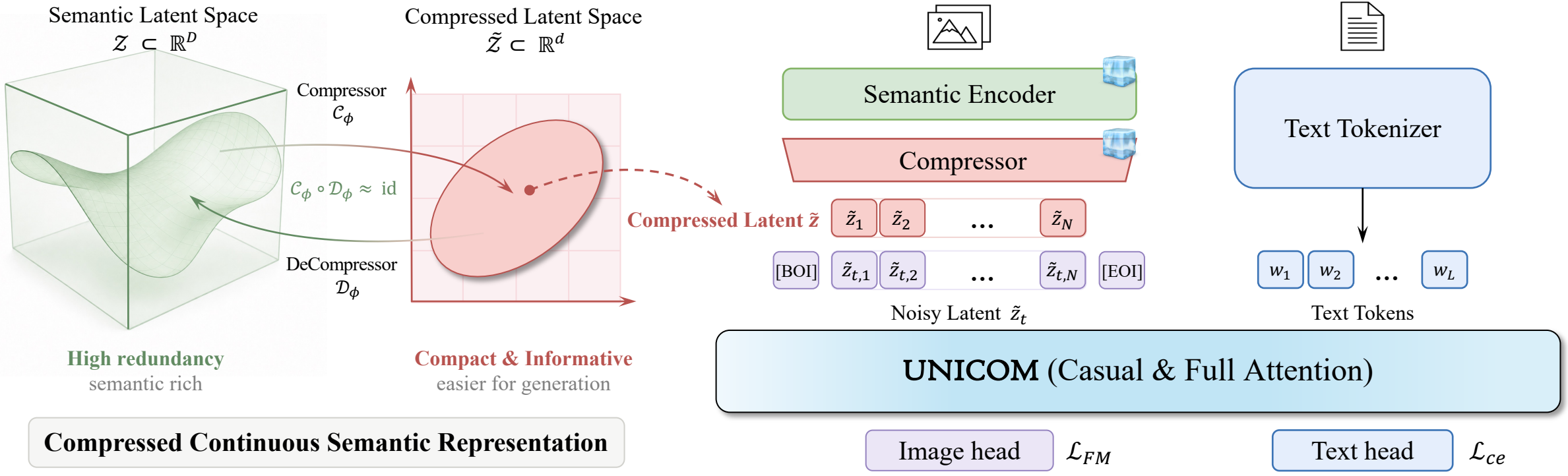

Semantic Latent Space
$\mathcal{Z} \subset \mathbb{R}^D$
Compressed Latent Space
$\tilde{\mathcal{Z}} \subset \mathbb{R}^d$
Compressor
$\mathcal{C}_\phi$
$\mathcal{C}_\phi \circ \mathcal{D}_\phi \approx$ id
DeCompressor
$\mathcal{D}_\phi$
High redundancy
semantic rich
Compact & Informative
easier for generation
Compressed Continuous Semantic Representation
Semantic Encoder
Compressor
Compressed Latent $\tilde{z}$
$\tilde{z}_1$ $\tilde{z}_2$ ... $\tilde{z}_N$
[BOI] $\tilde{z}_{t,1}$ $\tilde{z}_{t,2}$ ... $\tilde{z}_{t,N}$ [EOI]
Noisy Latent $\tilde{z}_t$
Text Tokenizer
w_1 w_2 ... w_L
Text Tokens
UNICOM (Casual & Full Attention)
Image head
$\mathcal{L}_{FM}$
Text head
$\mathcal{L}_{ce}$